SpaceX Falcon 9 Data Science Capstone

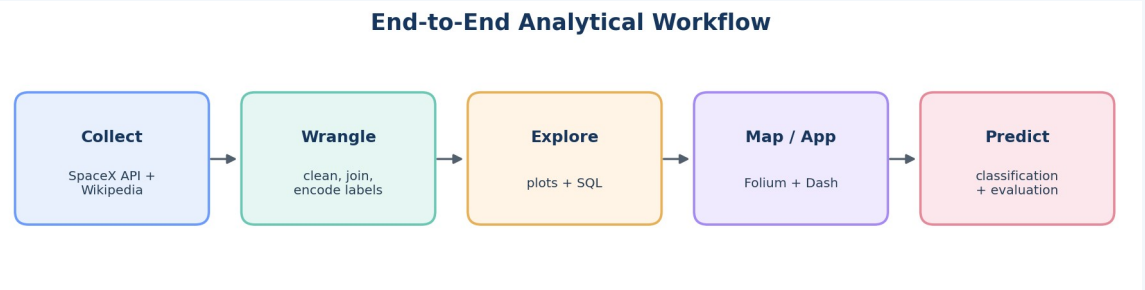
CAPSTONE PROJECT • HISTORICAL LAUNCH ANALYSIS

Exploring launch outcomes and predicting first-stage landing success

- Research question: Can historical launch, payload, orbit, launch-site and booster features help predict first-stage landing success?

- Workflow: API collection • web scraping • wrangling • EDA • SQL • Folium • Dash • classification

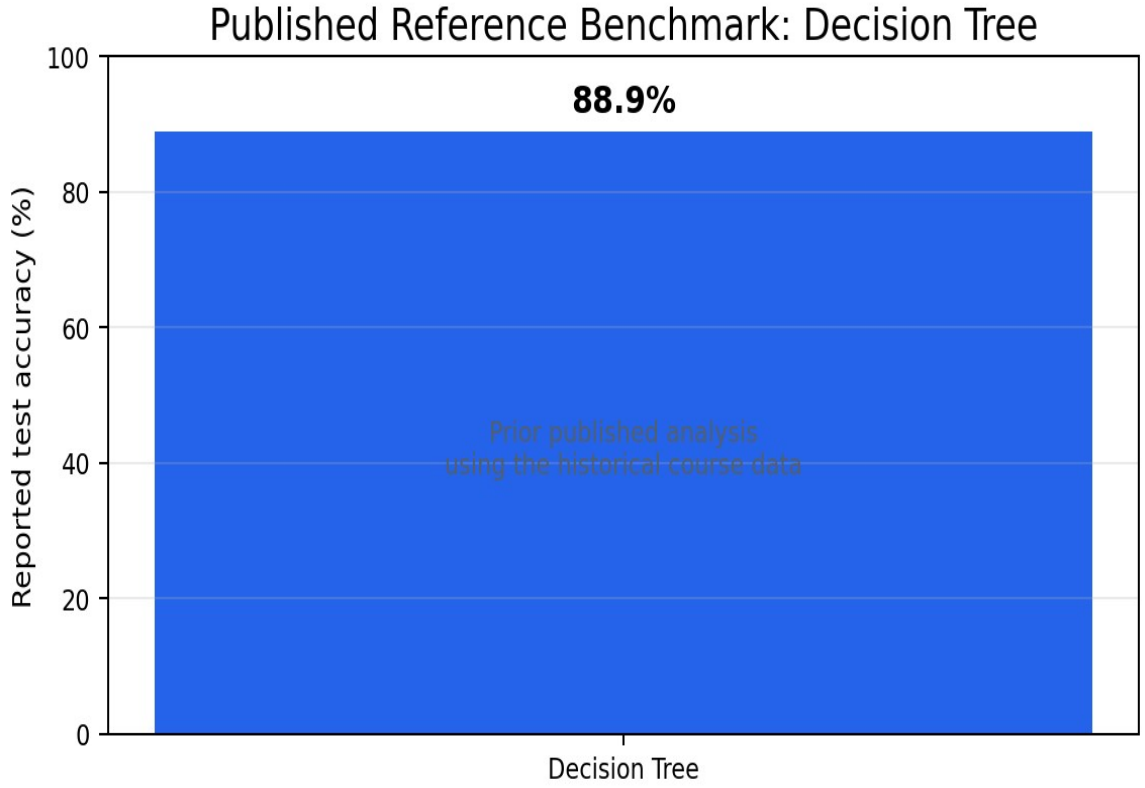
- Historical course dataset reference period: 2010–2020 in the published baseline study.



Data sources and benchmark findings are attributed in the References slide. Current outputs are regenerated by the included code.

Executive Summary

- The project combines two collection pathways (SpaceX API and launch-list web scraping) with a stable historical course dataset for reproducible evaluation.
- Published reference analysis found launch-site, payload-mass and orbit differences; it reported improving landing success over time.
- A prior analysis of this historical dataset reported a tuned Decision Tree accuracy of 88.9%. This is a reference benchmark, not a metric generated in this environment.
- The supplied pipeline regenerates the cleaned dataset, figures, SQL tables, Folium map, model metrics and confusion matrix when run.



Reference: Daniel Eduardo López, IBM-SpaceX capstone repository (2022). Recompute metrics on your own run before submission.

Introduction and Problem Statement

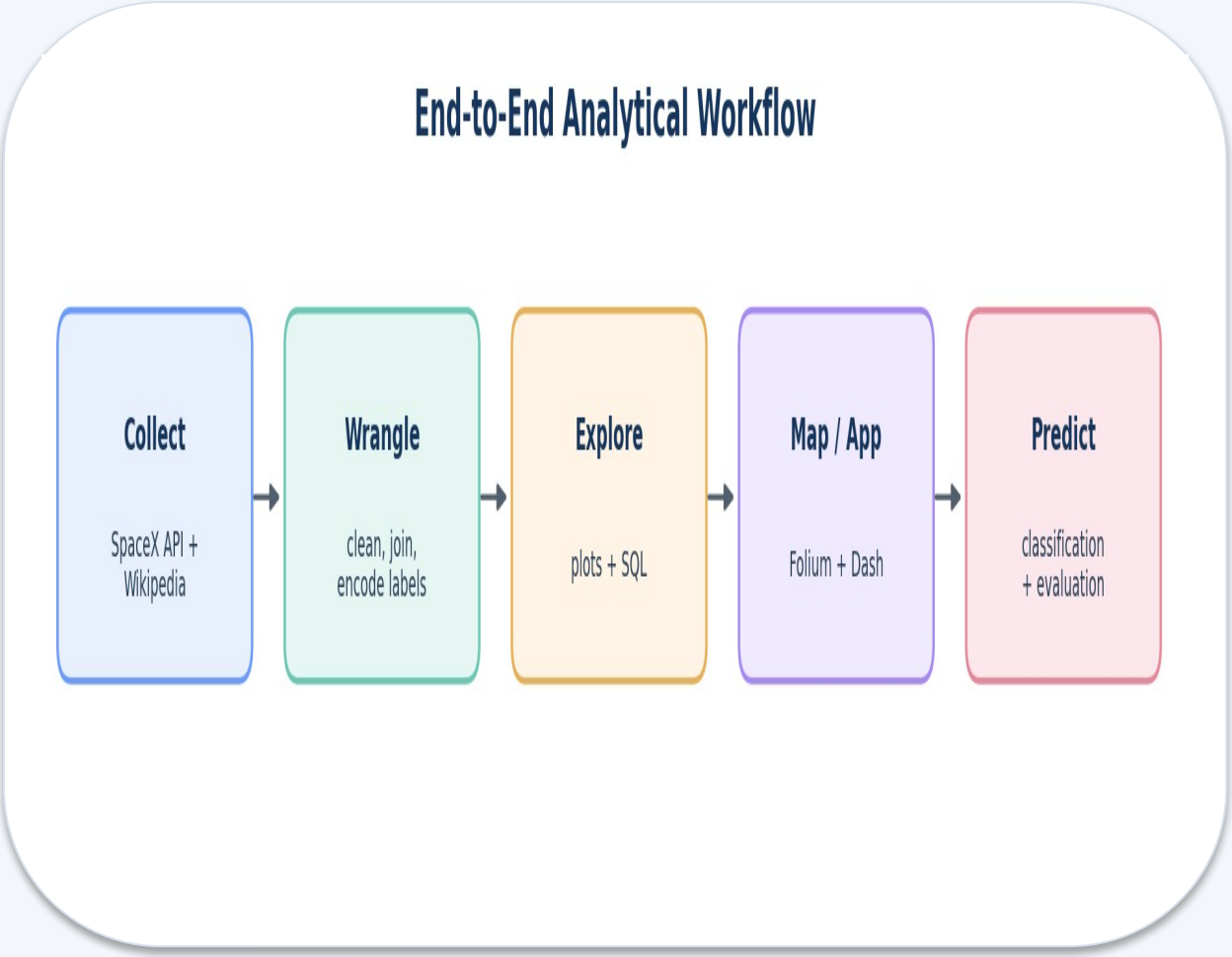
- Falcon 9 first-stage reuse is an important operational lever because successful recovery can enable refurbishment and reuse.
- Business question: Which historical mission features are associated with landing outcome, and how well can a classifier estimate landing success?
- Target: `Class` (0 = unsuccessful landing; 1 = successful landing) in the historical course dataset.
- Scope: descriptive analysis and prediction; associations do not establish that a feature causes a landing outcome.



Historical business context and research question follow the public capstone description; cited benchmark period ends in 2020.

Data Collection — SpaceX API

- Send GET request to `/v4/launches/past` and normalize JSON with `pandas.json_normalize`.
- Enrich launch IDs through `/rockets`, `/launchpads`, `/payloads` and `/cores` endpoints.
- Extract launch date, rocket/booster, payload mass and orbit, launch-site coordinates, recovery attempt and landing result where available.
- Quality controls: check HTTP status, handle missing fields and save the raw/normalized files for traceability.



API endpoint: https://api.spacexdata.com/v4/launches/past

Data Collection — Web Scraping

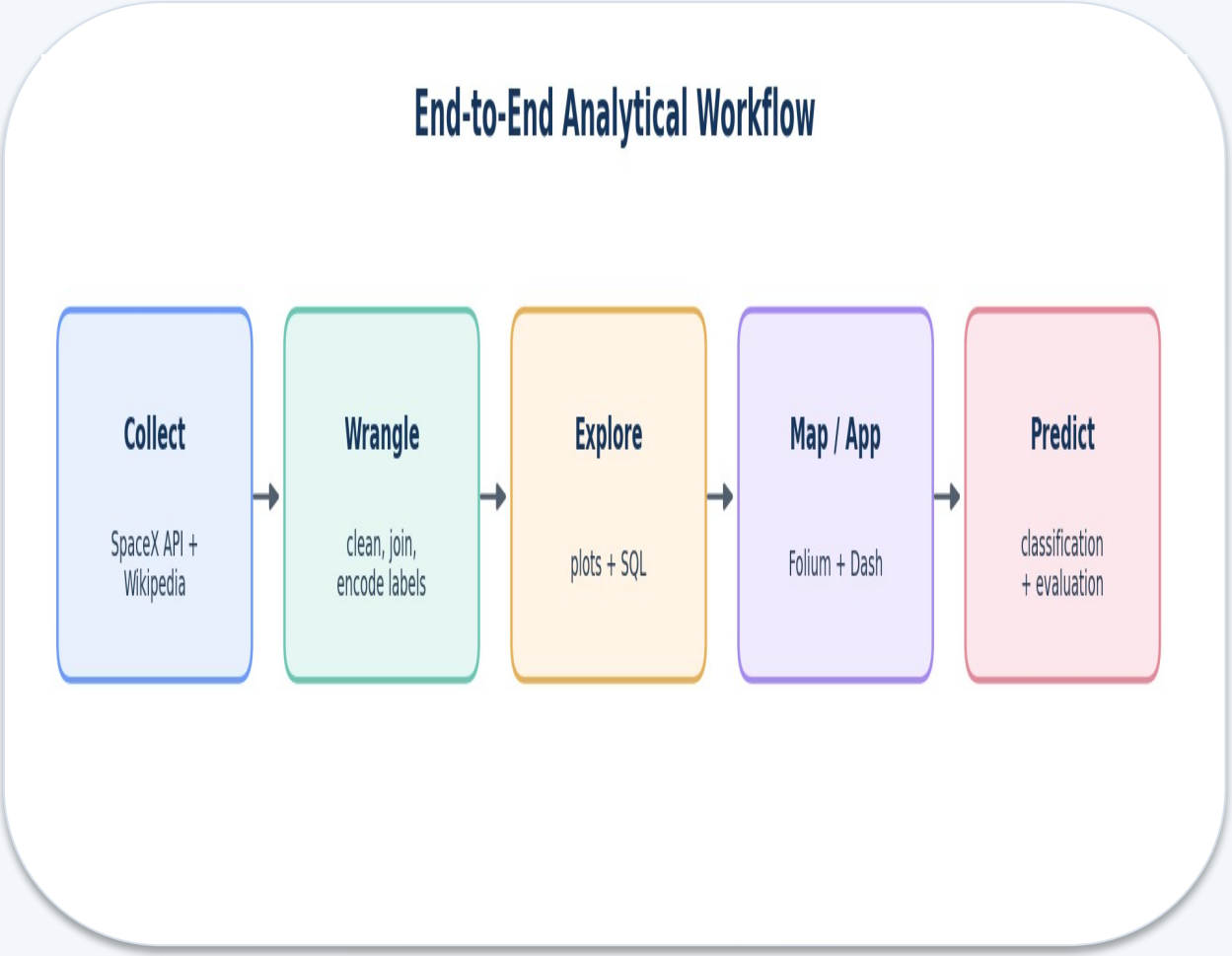
- Request the Wikipedia List of Falcon 9 and Falcon Heavy launches page using `requests`.
- Create a BeautifulSoup object, inspect `soup.title`, identify the launch table and parse it with `pandas.read_html`.
- Save the extracted table as a versioned CSV for inspection and compare schema changes before reuse.
- Web tables can change; the code checks status and records the table list rather than assuming a fixed table index.



Source: https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches

Data Wrangling Methodology

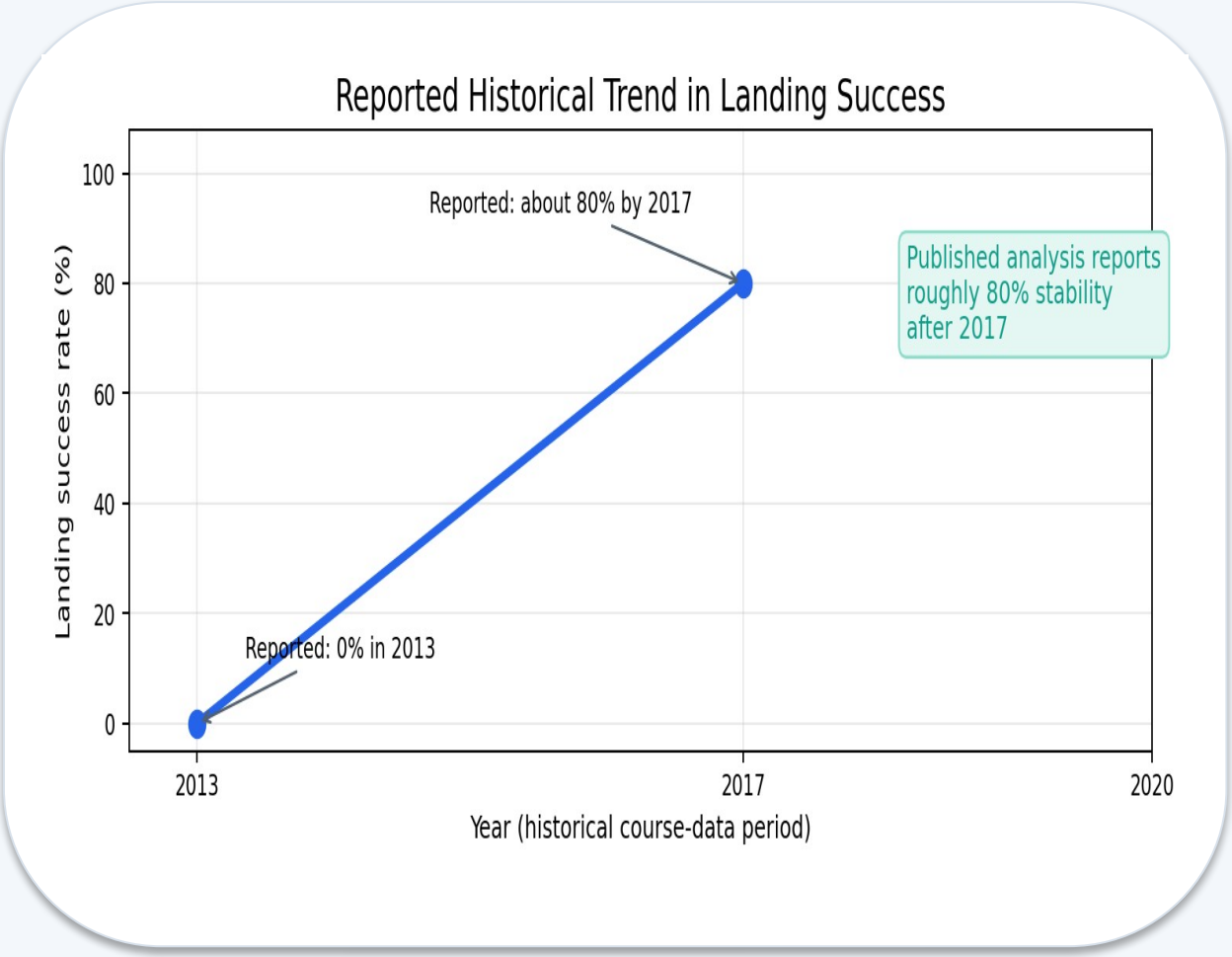
- Filter to Falcon 9 records for a consistent scope; standardize field names and types.
- Parse dates and create numeric model inputs; inspect missing values and duplicates before treatment.
- Impute missing predictor values inside a scikit-learn preprocessing pipeline.
- One-hot encode categorical fields and scale numeric values where appropriate. Split train/test data with stratification; keep preprocessing within the training pipeline to limit leakage.



The raw collector, historical-data loader, and model code are included in the ZIP.

EDA with Data Visualization

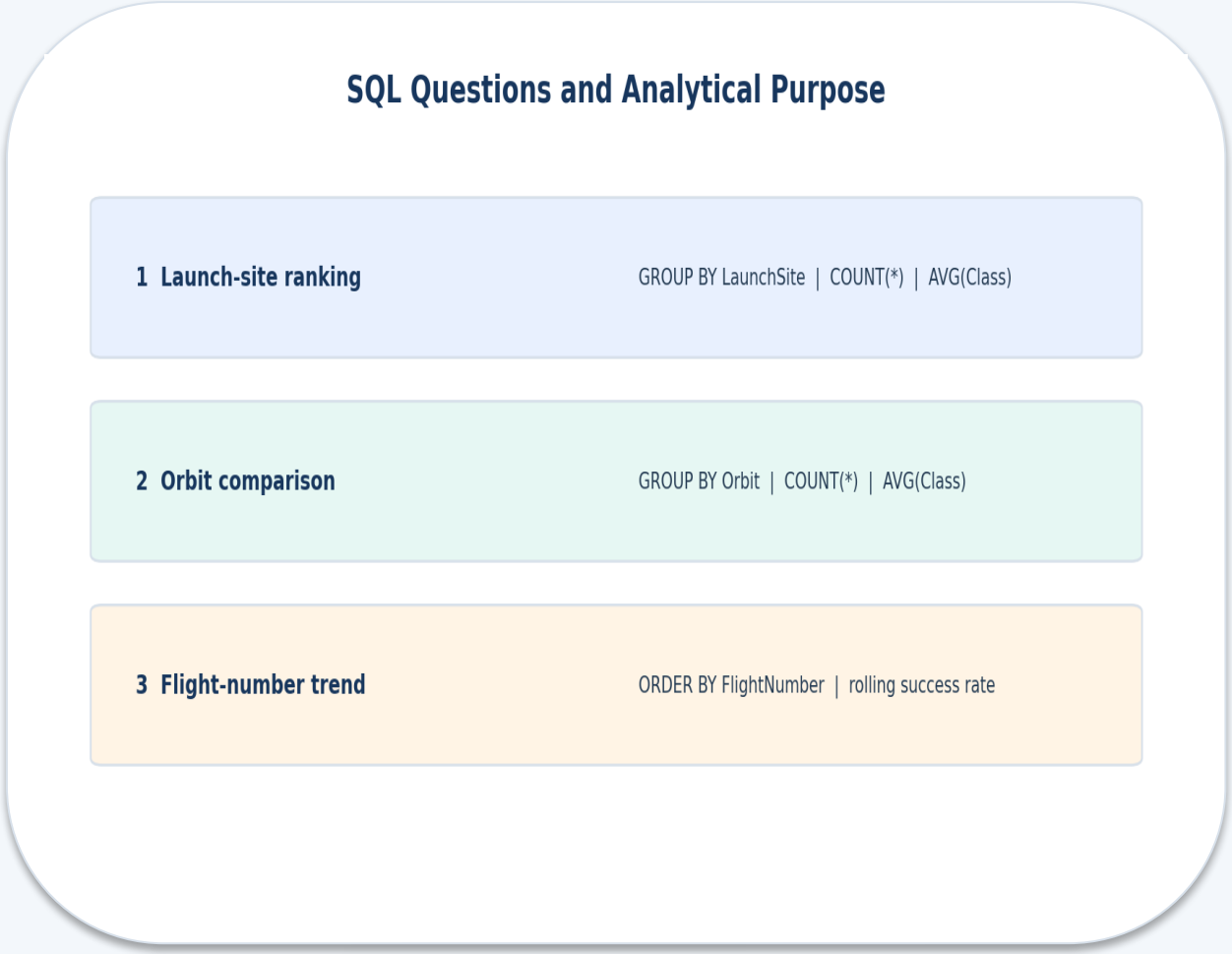
- Launch-site bar chart: compares sample size by site; it does not alone measure operational quality.
- Landing-outcome pie chart: visualizes target imbalance and explains why accuracy should not be the only metric.
- Payload scatter/strip plot: examines the distribution of payload mass across landing classes.
- Rolling success trend and orbit counts reveal temporal and mission-type patterns. The included code regenerates all figures from the loaded dataset.



The trend shown is the attributed published benchmark; use generated outputs/ figures for your own run.

EDA with SQL

- Load the cleaned launch-level data into SQLite so analysis is reproducible and queryable.
- Query 1: `GROUP BY LaunchSite` for record count and mean landing class.
- Query 2: `GROUP BY Orbit` for frequency and observed success rate; include counts so small groups are not overinterpreted.
- Query 3: order launches by flight number to examine trend changes. The script exports each SQL result as a CSV.



Generated files: sql_launch_site_summary.csv, sql_orbit_summary.csv, sql_flight_summary.csv.

EDA Result — Launch-Site Comparison

- Published baseline: CCAFS SLC-40 was the most frequently used launch site.
- VAFB SLC-4E was reported as the least used site, with comparatively few successful landings.
- KSC LC-39A was reported to have the highest count of successful missions and a 76.9% success ratio.
- Key insight: launch volume and success ratio answer different questions; do not infer the best site from count alone.

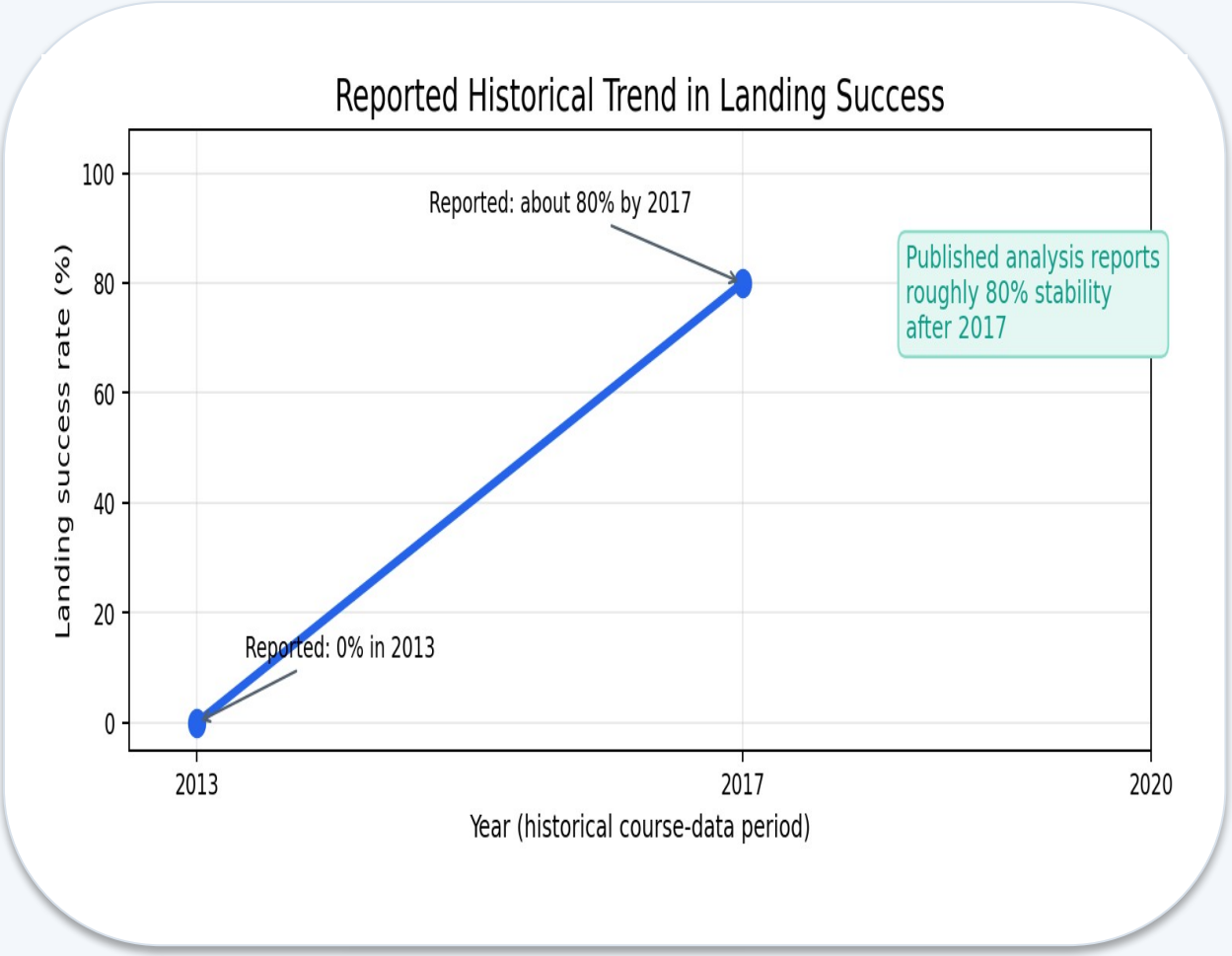
Launch-Site Comparison from the Referenced Historical Analysis

Launch site	Published historical finding
CCAFS SLC-40	Most frequently used launch site
KSC LC-39A	Highest number of successful missions; 76.9% reported success ratio
VAFB SLC-4E	Least frequently used launch site; low successful-landing count

All findings on this slide are reference findings from the published historical course-data analysis, not current SpaceX operational statistics.

EDA Result — Success Trend over Time

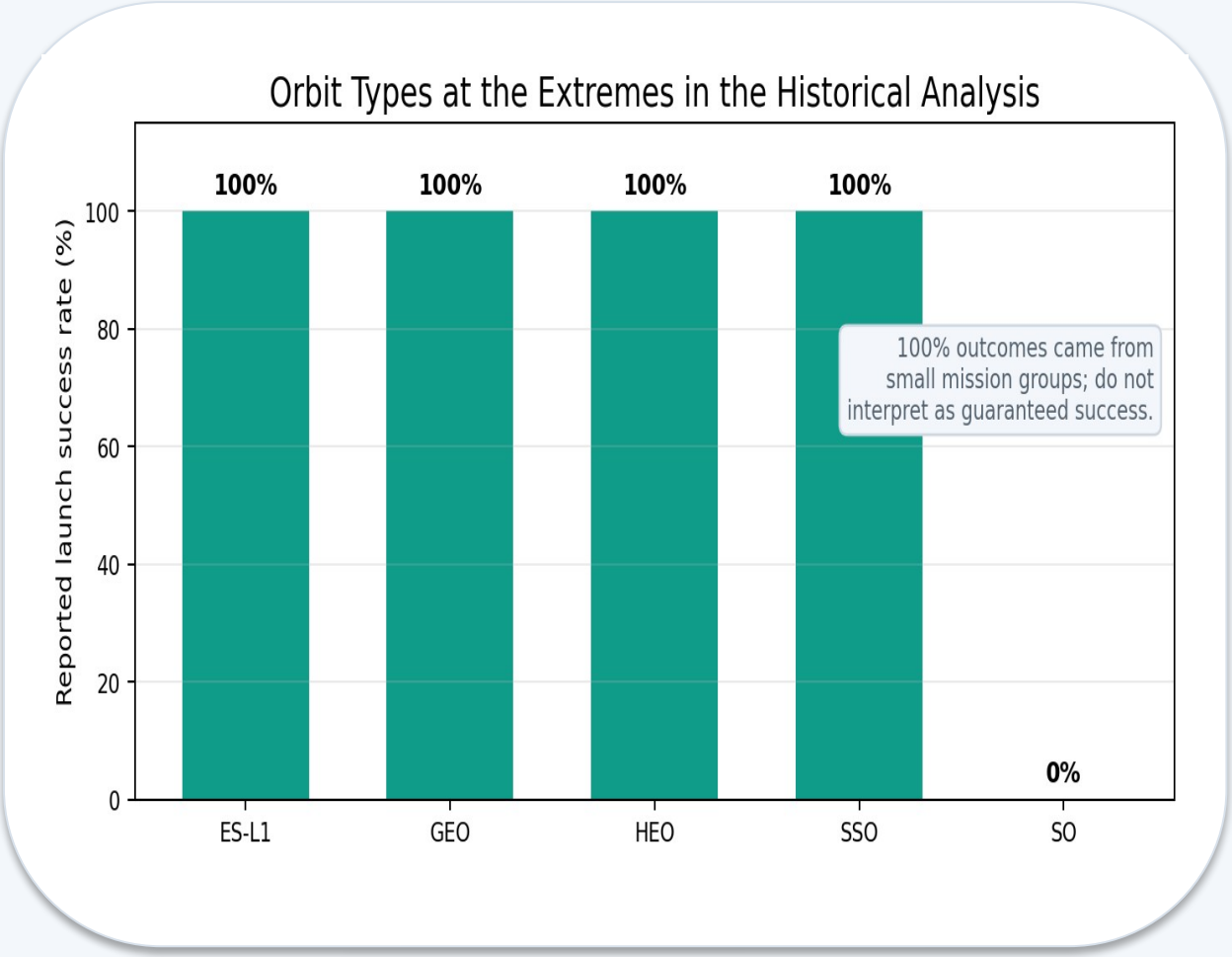
- The referenced study reported that landing success improved over the historical flight sequence.
- Its summary described success rising from 0% in 2013 to about 80% by 2017, then staying near 80%.
- The chart deliberately plots only the explicitly reported year/value points; the later plateau is described as a textual finding rather than inventing annual observations.
- The code computes a rolling success rate directly from the selected CSV for an updated result.



Reference values: published IBM-SpaceX capstone analysis, not an independently re-run analysis.

EDA Result — Payload and Orbit

- Reported high-success payload range: 1,952–5,300 kg. This is an observed range, not a causal threshold or guarantee.
- The reference analysis reported 100% observed success for ES-L1, GEO, HEO and SSO; the SO category was reported at 0%.
- The same analysis cautioned implicitly through small counts that 100% categories can be based on very few missions.
- The included code plots actual orbit counts and success rates with denominators, so small-sample effects can be assessed.



Use per-orbit counts alongside percentages when discussing uncertainty.

SQL Results — What the Tables Tell Us

- The historical reference analysis ranked CCAFS SLC-40 highest for launch usage and VAFB SLC-4E lowest.
- KSC LC-39A appeared as the site with the highest success count and 76.9% reported success ratio.
- Orbit summaries highlighted a mix of high observed success in a few low-frequency orbits and frequent mission types such as LEO, ISS, PO, GTO and VLEO.
- The run script exports the actual grouped SQL outputs from the course dataset so these observations can be checked quantitatively.

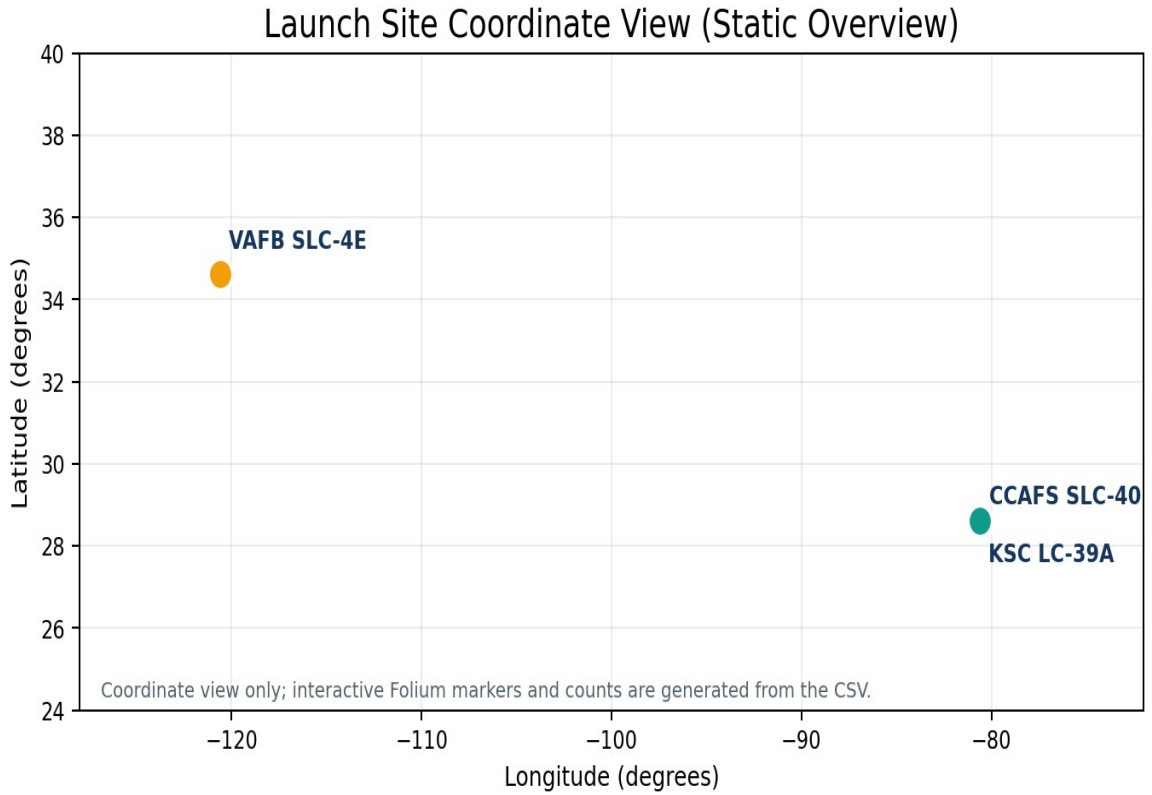
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SQL result CSVs are written beneath `outputs/` after execution.

Interactive Visual Analytics — Folium

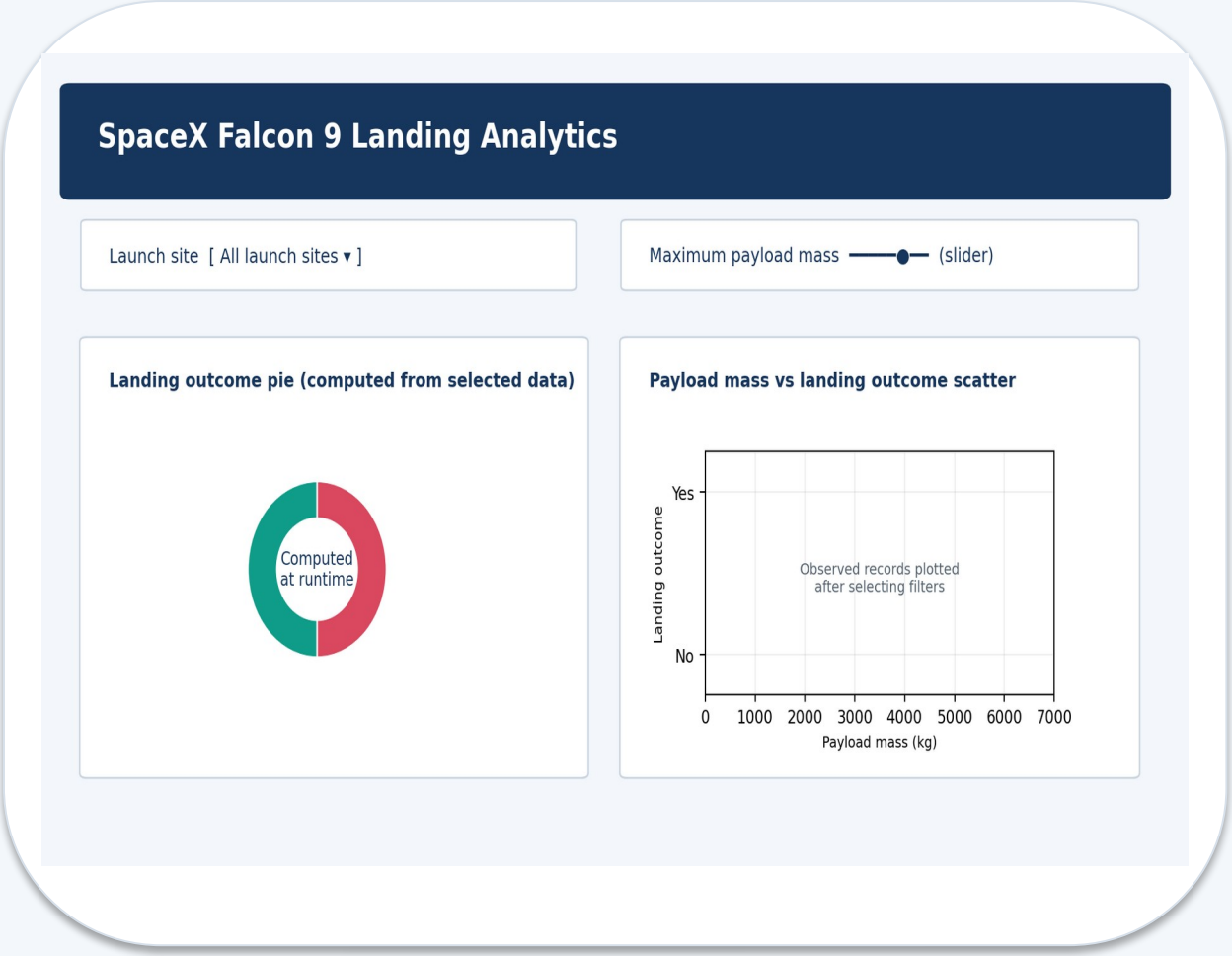
- The map uses site latitude/longitude from the dataset and puts one marker at each launch-site location.
- Marker popups show launch record count and observed landing success percentage.
- The script exports pairwise Haversine distances between launch-site coordinates for a reproducible proximity comparison. A published historical analysis reported KSC LC-39A at about 6.46 km from the Atlantic Sea and 72.21 km from Orlando; such distances depend on the chosen reference point.
- Run `python run_analysis.py` to create `outputs/spacex_launch_sites_map.html` and the pairwise distance CSV.



Actual interactive coordinate-based map is generated from the input CSV; this static view is only an overview.

Interactive Visual Analytics — Plotly Dash

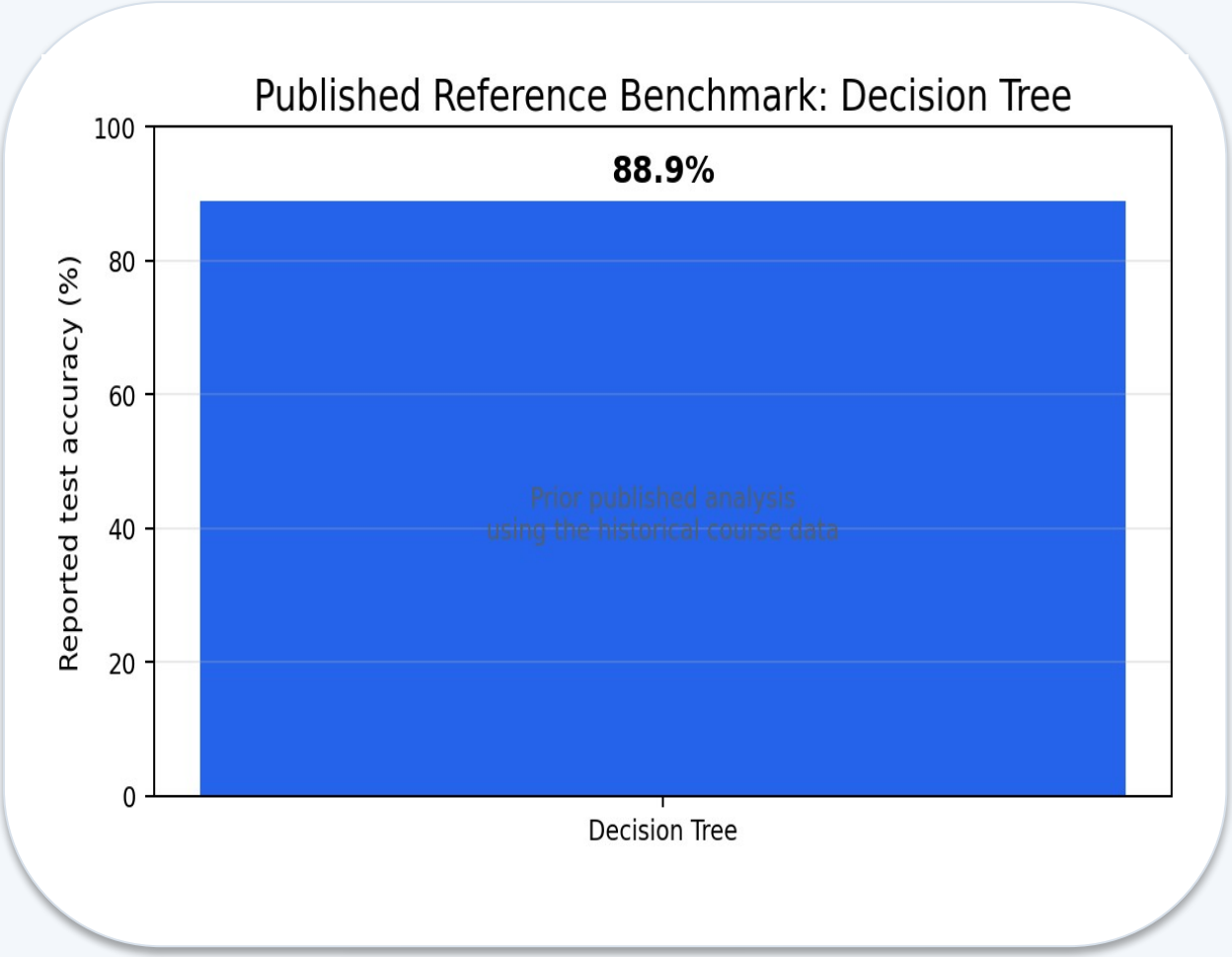
- The provided Dash app includes a launch-site dropdown and payload-mass slider.
- The pie chart dynamically summarizes successful versus unsuccessful landing outcomes for the selected site.
- The scatter plot maps observed payload mass against landing class and colors points by booster version or site.
- Run `python spacex_dash_app.py`, then open `http://127.0.0.1:8050/`. The slide image is a UI preview; live chart values come from the dataset.



Dashboard source file: spacex_dash_app.py. Use live screenshot from your own app if the course explicitly requires screenshots.

Predictive Analysis — Classification and Results

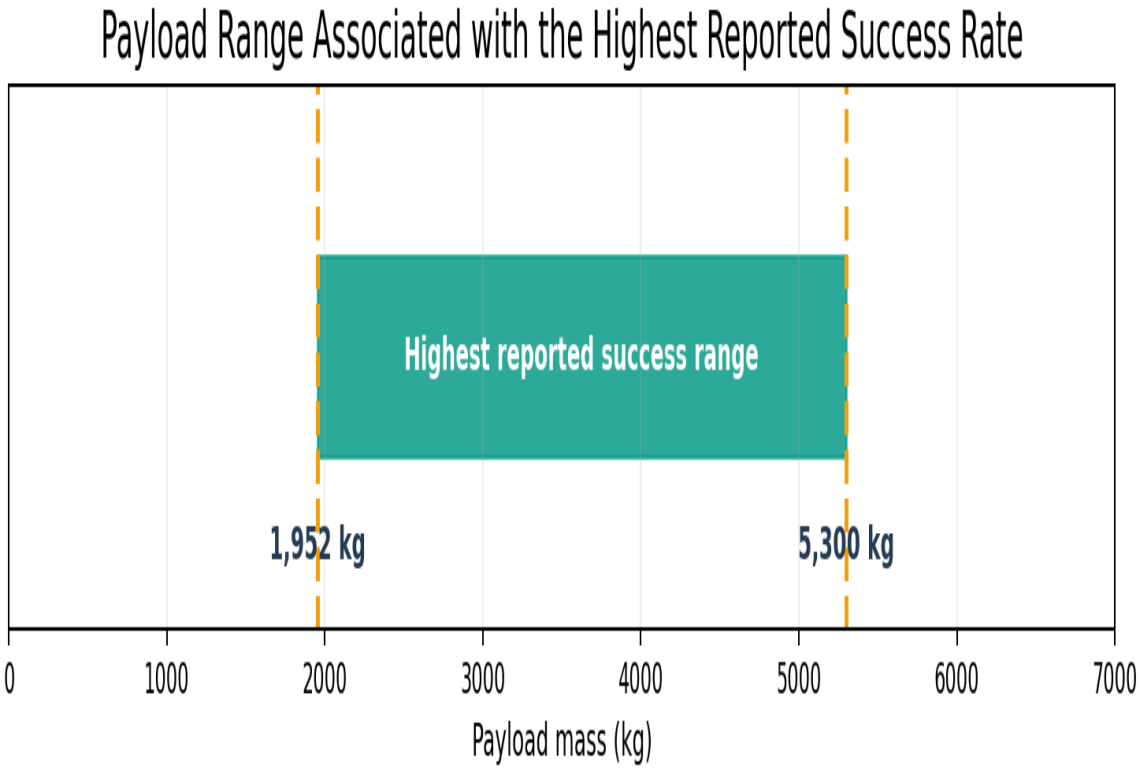
- Target: first-stage landing class. Candidate features include flight number, payload mass, orbit, site, booster version, reuse, grid fins, legs and block.
- Models compared by code: Logistic Regression, Decision Tree, Random Forest and Support Vector Machine.
- Published reference benchmark: a tuned Decision Tree reported 88.9% accuracy and performed better on successful than unsuccessful landings.
- Your code calculates current accuracy, precision, recall and F1 on a held-out test split; it saves a numeric confusion matrix and tuned tree parameters.



Do not report the 88.9% as your own run result unless your run reproduces it. Use outputs/model_metrics.csv.

Conclusion, Limitations and Innovative Insight

- Historical reference findings suggest landing success improved over the studied flight sequence and reached roughly 80% by 2017.
- Launch volume and success ratio should be analyzed separately: the busiest site was not the site reported with the highest success ratio.
- Orbit success percentages based on small samples can look perfect without being reliable; always display counts with rates.
- Limitations: historical cutoff, missing labels, class imbalance, changing operations and dataset-source drift. Re-run the pipeline and replace benchmarks with your measured outputs before final submission.



References: SpaceX API v4; Wikipedia launch list; Daniel Eduardo López, IBM-SpaceX capstone repository.